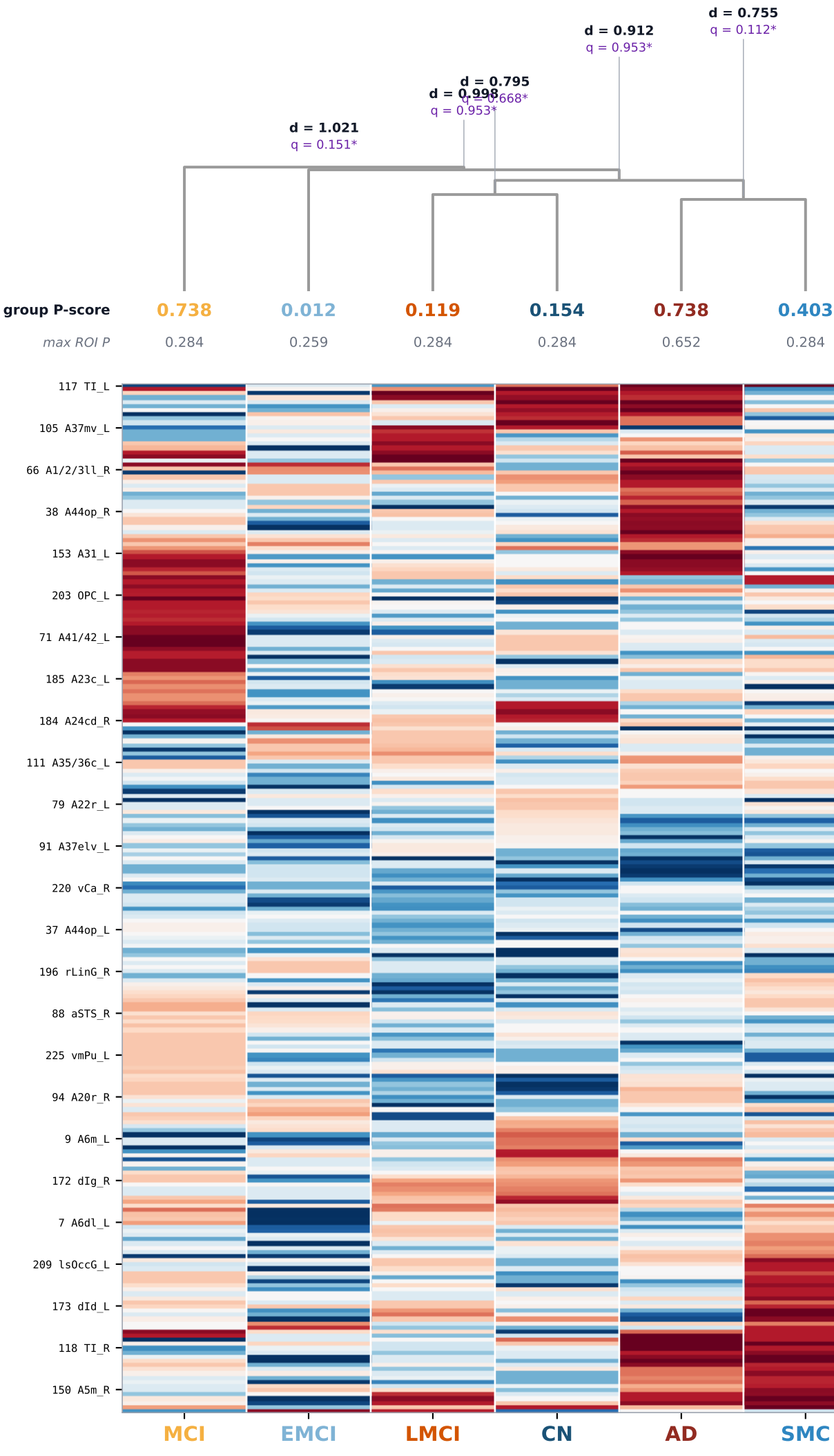
Hierarchical Clustering of Alzheimer's Diagnostic Groups
Regional amplitude of low-frequency fluctuation (mALFF)

cells are the ROI-level P-score = -log10(FDR-adjusted q) for each diagnostic group vs the pooled rest (BH over 246 x 6 = 1476 tests per biomarker)
groups clustered by correlation distance + average linkage over the 246-D P-SCORE profile
ROI rows ORDERED by euclidean + average on the same profiles (tree not drawn)



Brainnetome ROI (all 246 rows plotted, every 10th labelled, clustered order)

columns are ordered by the dendrogram above, NOT by disease stage

clinical progression order: CN → SMC → EMCI → MCI → LMCI → AD

bootstrap stability of the branches (1000 resamples of subjects within groups): {LMCI,AD} 46%; {CN,SMC,EMCI} 25%; {SMC,EMCI} 25%

group P-score = -log10(q) from the MULTIVARIATE stage-vs-rest permutation test (24-test family); max ROI P = largest ROI-level P-score, DESCRIPTIVE only

d = merge DISTANCE (geometric, not a p-value). *q = permutation q for that cluster contrast, SELECTIVE (clusters came from these data) -- anti-conservative, descriptive only.

COLOUR SHOWS RELATIVE VARIATION WITHIN THIS BIOMARKER ONLY, NOT PROXIMITY TO SIGNIFICANCE. Figures are NOT comparable to one another.

Descriptive only. Branch heights are distances, not p-values. No group reaches q < 0.05 and no ROI x stage test in this project does either.